

Course Code: CSE-2422
Course Title: Computer Algorithm Lab
Credit Hours: 1
Contact Hours: 2 lab hours per week
Type: Core, Engineering
Prerequisite: CSE-2322 (Data Structures Lab)
Co-requisite: CSE-2421 (Computer Algorithms)

Instructor's Name and Office Location

Instructor: Mysha Sarin Kabisha

Assistant Lecturer, Dept. of CSE, IIUC

Office Location: Room # 406, Female Academic Building - 2

Contact: 01858759047

Email : myshasarincse@iiuc.ac.bd / myshasarin16@gmail.com

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	20Marks
	SEE: Semester End Examination		60 Marks

Laboratory works based on CSE-2421

Course Objective: The objectives of this course are to

1. familiarize with the major algorithms and data structures
2. proficiently apply important algorithmic design paradigms and methods of analysis
3. synthesize efficient algorithms in common engineering design situations

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Demonstrate a familiarity with major algorithms and data structures	50%
2.	Apply important algorithmic design paradigms and methods of analysis	30%
3.	Synthesize efficient algorithms in common engineering design situations	20%

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Demonstrate a familiarity with major algorithms and data structures	PLO1	Cognitive/Apply	Lecture, Class discussion, Assignment, Lab work, Note, Competitive Problem Solving	Exam, Quiz, Assignment, Competitive Problem Solving

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO2	Apply important algorithmic design paradigms and methods of analysis	PLO1 PLO2	Cognitive/Apply	Lecture, Class discussion, Assignment, Lab work, Note, Competitive Problem Solving	Exam, Quiz, Assignment, Competitive Problem Solving
CLO3	Synthesize efficient algorithms in common engineering design situations	PLO2	Cognitive/Create	Lecture, Class discussion, Assignment, Lab work, Note, Competitive Problem Solving	Exam, Quiz, Assignment, Competitive Problem Solving

Experiment Names:

Implementation and Analysis of Insertion sort, Merge sort, Quick sort, Heap sort, Counting sort, Radix sort, Matrix Chain Multiplication, Longest Common Subsequence, Job selection problem, Breadth First Search, Depth First Search, Bellman-Ford's Algorithm, Dijkstra's Algorithm, Floyd-Warshall Algorithm, Random Number Generation, Backtracking Algorithm, N-Queen Problem, Huffman Algorithm, Knapsack Problem, Minimum Spanning Tree.

To perform also other experiments relevant to this course.

Course Assessment Pattern:

Bloom's taxonomy domain/level	Evaluation out of 100 marks			
	CIE (50 marks)		SEE (50 marks)	Total (100)
	Attendance (10)	Continuous Assessment (40)	Final Exam (50)	Total (100)
Cognitive/Remember	-	-	-	-
Cognitive/Understand	-	-	-	-
Cognitive/Apply	-	10	10	20
Cognitive/Analyze	-	10	10	20
Cognitive/Evaluation	-	10	10	20
Cognitive/Create	-	10	20	30
Affective/Responding	10	-	-	10

Text Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, 3rd Edition, MIT Press, 2009, ISBN-13: 978-0262033848
2. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Fundamentals of Computer Algorithms, 1st Edition, Orient Black Swan, 2008, ISBN-13: 978-8173716126

Reference Books:

3. Robert Sedgewick, Algorithms, 4th Edition, Addison-Wesley Professional, 2011, ISBN-13: 978-0321573513

4. Steve Skiena, The Algorithm Design Manual, 2nd Edition, Springer, 2008, ISBN-13: 978-1849967204
5. Robert L. Kruse, Alexander J. Ryba, Data Structures and Program Design in C++, 1st Edition, Prentice Hall, 1998, ISBN-13: 978-0137689958
6. Mark Allen Weiss, Data Structure and Algorithm analysis in C++, 4th Edition, Pearson, 2013, ISBN-13: 978-0132847377
7. Jeff Edmonds, How to Think About Algorithms, 1st Edition, Cambridge University Press, 2014, ISBN-13: 978-1107439887
8. Steven Halim and Felix Halim, Competitive Programming 3, 3rd Edition, Lulu.com, 2013
9. Donald Ervin Knuth, The Art of Computer Programming, Volume 1, Addison-Wesley Professional, 3rd edition, 1997, ISBN-13: 978-0201896831
10. Donald Ervin Knuth, The Art of Computer Programming, Volumes 1, 2, 3, 4A, Addison-Wesley Professional, 1st edition, 2011, ISBN-13: 978-0321751041

Online Resources:

I) <https://www.topcoder.com/community/competitive-programming/tutorials/>

Grading Policy:

As per IIUC grading policy

Weekly Activity Plan and List of Experiments:

Week	Activities	Experiments
1	Lab work	1. Implementation of Insertion-sort 2. Implementation of Merge-sort
2	Lab work	3. Implementation of Quick-sort with its randomized
3	Lab work	4. Implementation of Heap and Heap-sort 5. Implementation of Priority queue using binary heap
4	Lab work	6. Implementation of Counting sort 7. Implementation of Radix sort
5	Lab work	8. Solving Matrix-chain multiplication problem 9. Solving longest common subsequence problem
6	Lab work	10. Solving problem with the technique of memorization 11. Solving selected competitive programming problem that requires dynamic programming
7	Lab work	12. Solving activity selection problem 13. Implement Huffman tree and generating prefix
8	Lab work	14. Implementation of Naive string-matching algorithm 15. Implementation of Rabin-Karp algorithm
9	Lab work	16. Implementation of Breadth First Search 17. Implementation of Depth First Search
10	Lab work	18. Implementation of Kruskal's Algorithm for finding minimum spanning tree 19. Implementation of Prim's Algorithm for finding minimum spanning tree
11	Lab work	20. Implementation of Dijkstra's algorithm for solving single-source shortest path problem 21. Implementation of Bellman-Ford's algorithm for solving single-source shortest path problem

Week	Activities	Experiments
12	Lab work	22. Implementation of Floyd-Warshall's algorithm for solving all-pairs shortest path problem
13	Lab work	23. Determining whether two line segment intersect 24. Determining convex hull of a set of points using Graham's scan algorithm
14	Lab work	25. Implementation of extended Euclid's algorithm for finding GCD 26. Implementation of different prime number generation algorithms 27. Solving N-Queen Problem 28. Solving different backtracking problems 29. Write a program to find the value of $X^n \bmod m$
15	Programming Contest	Testing the problem solving skills of students by giving them problems